

# Tutorial

## Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.  
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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## WEEK II

## 4. STRUCTURE OF THE ATOM

- **Planetary model for light ions / Ex. 4.16**

Calculate the speed and radial acceleration for a ground-state electron in the hydrogen atom H. Do the same for the  $\text{He}^+$  and the  $\text{Li}^{++}$  ions.

- **Angular momentum in Bohr's hydrogen atom model / Ex. 4.20**

(a) Calculate the angular momentum in  $\text{kgm}^2/\text{s}$  for the lowest electron orbit in the hydrogen atom. Compare the result with Planck's constant  $h$ . (b) Repeat for an electron in the  $n$ -th state of hydrogen.

- **Binding energy of light ions / Ex. 4.22**

What is the total mechanical energy for a ground-state electron in H,  $\text{He}^+$  and the  $\text{Li}^{++}$  atoms? For which atom is the electron most strongly bound? Why?

- **Absorption transition / Ex. 4.23**

A hydrogen atom in an excited state absorbs a photon of wavelength  $\lambda = 410 \text{ nm}$ . What were the initial and final states of the hydrogen atom? Use Rydberg constant in the infinite nucleus mass approximation:  $R_\infty = 1.097\,373 \times 10^7 \text{ m}^{-1}$ .

- **Emission transition / Ex. 4.24**

A hydrogen atom in an excited state emits a photon of wavelength  $\lambda = 95 \text{ nm}$ . What are the initial and final states of the hydrogen atom? Use Rydberg constant in the infinite nucleus mass approximation:  $R_\infty = 1.097\,373 \times 10^7 \text{ m}^{-1}$ .

- **Isotope shift of the Balmer series / Ex. 4.26**

The *isotope shift* of spectral lines refers to the shift in wavelengths (or frequencies) due to the different isotopic masses of given elements. Find the isotope shifts in wavelength for deuterium and tritium compared with hydrogen for each of the four visible lines of the Balmer series.

- **Muonic atom / Ex. 4.31**

A muonic atom consists of a muon (mass  $m_\mu = 106 \text{ MeV}/c^2$  and charge  $q = -|e|$ ) in place of an electron. For the muon in a hydrogen atom, what is (a) the smallest radius and (b) the binding energy of the muon in the ground state? (c) Calculate the series limit of the wavelength for the first three series.

- **Positronium / Ex. 4.32**

Positronium is an atom composed of an electron and a positron (mass  $m_{e^+} = m_e$ , charge  $q = |e|$ ). Calculate (a) the distance between the particles and (b) the energy of the lowest energy state of positronium. *Hint*: What is the reduced mass of the two particles?

- **X-ray spectra of Pb and Bi / Ex. 4.39**

If the resolution of a spectrograph is  $\Delta\lambda = 1 \times 10^{-12} \text{ m}$ , would it be able to separate the  $K_\alpha$  lines for lead (Pb) and bismuth (Bi)? Explain.

- **Redshifted  $K_\alpha$  and  $K_\beta$  lines of hydrogen / Ex. 4.50**

In the early 1960s the strange optical emission lines from starlike objects that also produced tremendous radio signals confused scientists. Finally, in 1963 Maarten Schmidt of Palomar Observatory discovered that the optical spectra were just those of hydrogen but redshifted because of the tremendous velocity of the object with respect to Earth. The object was moving away from Earth at a speed of  $50.000 \text{ km/s}$ ! Compare the wavelengths of the normal and redshifted spectral lines. for the  $K_\alpha$  and  $K_\beta$  lines of the hydrogen atom.

## 5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

- **Bragg scattering off NaCl crystal / Edited from Example 5.1 + Ex. 5.2**

X-rays scattered from rock salt (NaCl) are observed to have a first maximum ( $n = 1$ ) at an angle of  $\phi = 21.3^\circ$  from the incident direction. (a) What must be the wavelength  $\lambda$  of the incident radiation? (b) What is the angular separation between first and second-order diffraction peaks? Assume that the scattering planes that are parallel to the surface of the crystal.

- **Comparing de Broglie wavelength for atomic and biological matter / Ex. 5.17**

Calculate the de Broglie wavelength of (a) an oxygen ( $\text{O}_2$ ) molecule darting around the room at  $v = 480 \text{ m/s}$  and (b) an *Escherichia coli* bacterium of mass  $m_b = 6.5 \times 10^{-13} \text{ kg}$  which has been measured to move at a speed of  $v_b = 1.0 \times 10^{-5} \text{ m/s}$ . *Optional:* Compare the de Broglie wavelength and the object's size in each case.

- **De Broglie wavelength of a rock / Ex. 5.5**

Calculate the de Broglie wavelength of a rock with mass  $m = 1.2 \text{ kg}$  thrown with a speed  $v = 6.0 \text{ m/s}$  into a pond. Is this wavelength similar to that of the water waves produced? Explain.

- **Spatial resolution of TEM / Ex. 5.7**

Transmission electron microscopes that use high-energy electrons accelerated over a range of voltages from  $\Delta V = 40 \text{ kV}$  to  $\Delta V = 100 \text{ kV}$  are employed in many applications including the study of biological samples (for example, a virus) and nanoscience research and development (alloy particles and carbon nanotubes, for example). What would be the spatial limitation for this range of electrons? It is often true that resolution is limited by the optics of the lens system, not by the intrinsic limitation due to the de Broglie wavelength.

- **Phase and group velocity / Ex. 5.24**

Consider electrons of kinetic energy  $K = 6.0 \text{ eV}$ . Find the de Broglie wavelength  $\lambda$ , particle speed  $v$ , phase velocity  $v_{\text{ph}}$ , and group velocity  $u_{\text{gr}}$ .